Prototype V1

CS1710 · 64 Tomorrows · WeatherNext 3 forecast, New England

1. Team members

Students who worked on this prototype: **Henry Mattoni**, **Ryan Eto**, **Luke Tan**.

Confirm before submitting that each of the three actively worked on this submission. The contribution split is not recorded in the repository.

2. What changed since Milestone 7

The Milestone 7 storyboard set the story: a hook (the single number a weather app shows), rising insights (how uncertainty grows and varies), a main message (the app picks one future and hides how unsure it is), and a solution (learn when to trust a forecast). Prototype V1 rebuilds the website around that arc. The page is now organized into four acts plus an Explore section, and the new hook view is the novel visualization for this milestone.

- **Act 1 · Hook.** New view: *one number, many possible tomorrows* (`fanHook`).
- **Act 2 · Rising insights.** Two days ahead (temperature band), how fast the gap grows by place and variable (`rangeGrowth`), where the fog thickens (`fogMap`), and whether forecasts jump between runs (`timeMachine`).
- **Act 3 · Main message.** "Your app picks one future": forecast roulette, the freezing-precipitation yes-or-no question, and the activity map.
- **Act 4 · Solution.** A short rule of thumb by lead time, then the trust-horizon map and the route planner, with links to the two dashboards.
- **Explore.** Cloud cover, wind, and the storm cone, kept as optional depth.

3. Complete data pipeline

The pipeline is finished and produces the data the site uses. Source: Google WeatherNext 3 ensemble statistics (mean and 10th to 90th percentiles of the 64 members) from `gs://weathernext3_statistics_spatial`.

- `pipeline/fetch_wn3_stats.py` streams the latest 6-hourly run and crops it to New England.
- `pipeline/fetch_time_machine.py` pulls the same valid window from 10 earlier runs (for the time machine).
- `pipeline/fetch_storm.py` pulls wide-domain sea-level pressure (for the storm cone).

- `pipeline/build_web_data.py` and `pipeline/build_extras.py` write the compact site data: 44 field files stored as int16 binaries, per-town time series for 12 places, and the geography files.

Current data: run initialized 2026-10-07 12Z, 88 lead times from +1 h to +168 h (hourly to 48 h, then 3-hourly), 12 towns. Re-running the fetch streams about 63 GB, so it is documented but not re-run for this submission.

4. Two functional D3 visualizations

4a. Novel view: one number, many possible tomorrows (new)

This is the hook of the story. The left card shows what a weather app shows: one temperature. The right chart is the same hour across all 64 forecasts, drawn as a fan of percentile bands. The card's number is a point on that fan. Its three answers, 1-in-10 colder, most likely, and 1-in-10 warmer, are plotted as dots on the scrub line.

- **Data loading:** `manifest.json` for the valid times, and `points.json` for the selected town's 10th, 25th, 50th, 75th and 90th percentile temperatures.
- **Rendering:** D3 area bands (10–90 and 25–75), a median line, and dashed 10th and 90th percentile edges. The chart is re-rendered on resize and on theme change.
- **Interaction:** move the pointer across the chart to pick a time; the card updates. Tap or press Enter on the card to step through the three answers. The active answer is highlighted in the card and on the chart.

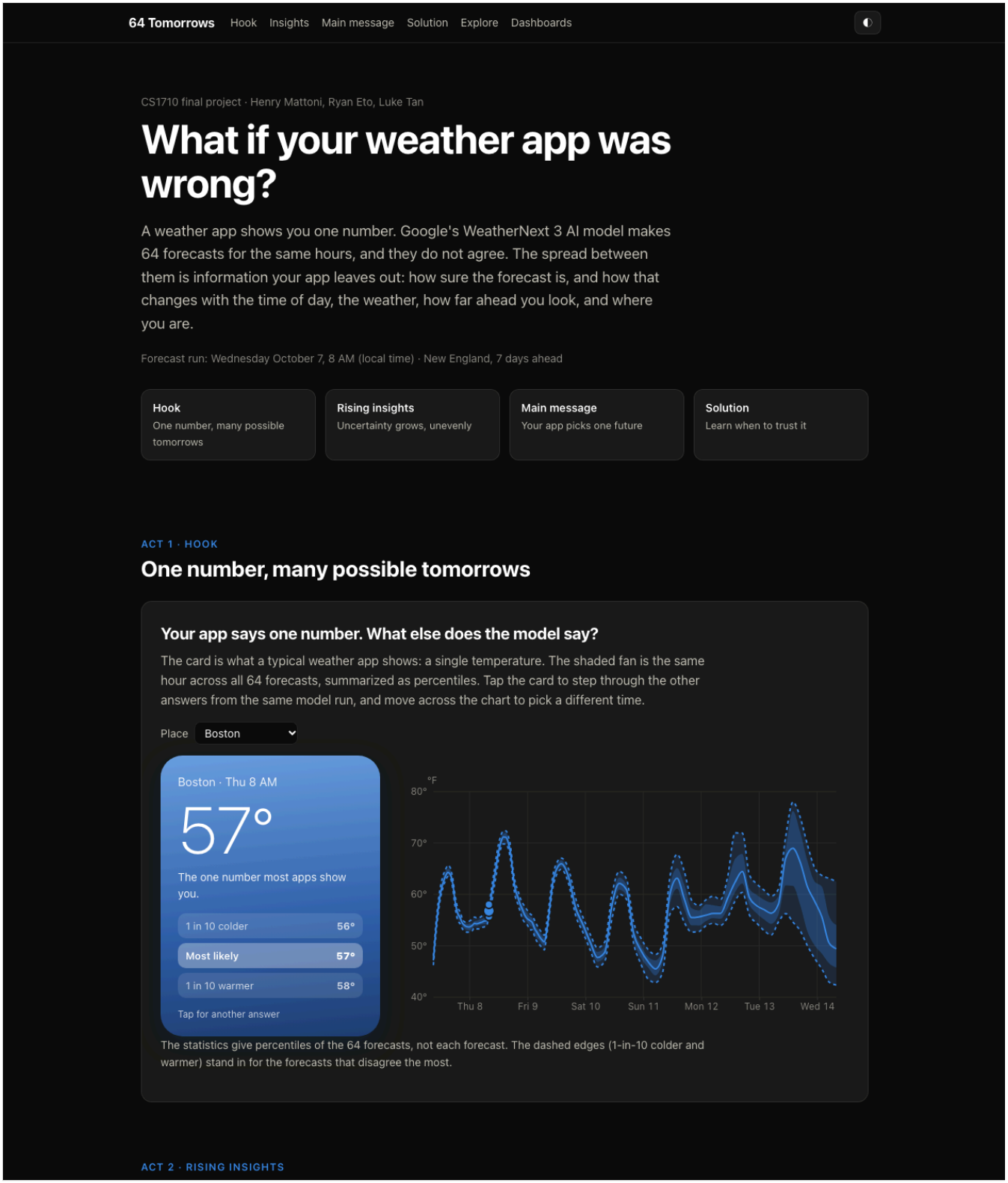


Figure 1. Prototype V1 hook view, Boston, dark theme, desktop width. Captured from the local build.

Known limit: the statistics give percentiles, not each of the 64 forecasts. The dashed edges are the 10th and 90th percentiles, so they stand in for the "far-end members" in the storyboard, not for real members. The card is also not a copy of any particular app's design.

4b. Rising insights: how fast does the gap grow, and where? (reused, reframed)

This is the existing growth chart (also the core of the Milestone 6 uncertainty dashboard), now placed in Act 2. It plots the gap between the 1-in-10 high and low for four places (Boston, Pittsfield, Cape Cod, Mt Washington) over three variables (temperature, wind, rain rate). Its data loading and rendering are unchanged. The copy now names the jump that shows up in the data (see Section 6).

5. Website structure

The page follows the storyboard's order. A sticky header links to each act. Each section is a card with a title stated as a question or claim, a one-line explanation, the control, and the chart. The views load lazily as they scroll into view. At phone width the hook card stacks above its chart and the story map becomes a two-column grid. Desktop and phone renders were checked in headless Chrome. The phone check was cropped, so it is not yet verified at a true 390 px width.

6. Insights checked against the data

Each storyboard claim was checked against the statistics before it went into the site copy.

Storyboard claim	What the data shows (this run)	Status
The first couple of days are very reliable.	Boston's 10–90 gap is 1.3 °C at +24 h and 1.4 °C at +48 h, against 3.1 °C at +96 h. The 12-town mean is 1.3 °C at +24 h.	Supported
Uncertainty is largest further away in time.	Boston: 1.3 °C (+24 h), 2.1 °C (+72 h), 3.1 °C (+96 h), 11.2 °C (+168 h).	Supported
When uncertainty grows it jumps instead of sloping up.	Boston's gap moves from 3.6 °C (+144 h) to 13.0 °C (+150 h), with the median moving from 14.5 °C to 20.5 °C. That is a clear jump. In the first 48 h, however, the top 10% of hourly steps account for only 26% of the total change, so the jumping is concentrated in the late forecast here, not the early one.	Partly supported
Uncertainty follows a daily pattern.	Averaged over the 12 towns for the first 48 h, the gap is lowest at 09:00 local (1.27 °C) and highest at 01:00 (1.74 °C). The swing is about half a degree.	Supported, small
Coast is less uncertain than inland.	First 48 h: coastal towns 1.45 °C, inland 1.64 °C. +48 to +120 h: 2.69 °C against 2.96 °C. The difference is small and comes from one run.	Weakly supported; worded as "slightly narrower"
Different variables have different patterns.	At +24 h the wind gap ranges from 1.2 to 2.3 across towns. The rain-rate gap ranges from 0 to 1.8 mm/h, so rain is much more uncertain in some places.	Supported

Some areas are less certain than others.	Mt Washington's gap is 1.6 °C at +24 h and 7.9 °C at +144 h. Cape Cod's is 0.8 °C at +24 h.	Supported

The earlier "top 10% of steps carry 42–56% of the change" figure was an error. It mixed hourly and 3-hourly steps after +48 h. The corrected 26% figure is used here.

7. Draft designs (not yet implemented)

Three visualizations are designed but not in the site. Each draft uses real numbers from this run, drawn with `docs/prototype/make_draft_figures.py`.

Draft A · Daily rhythm

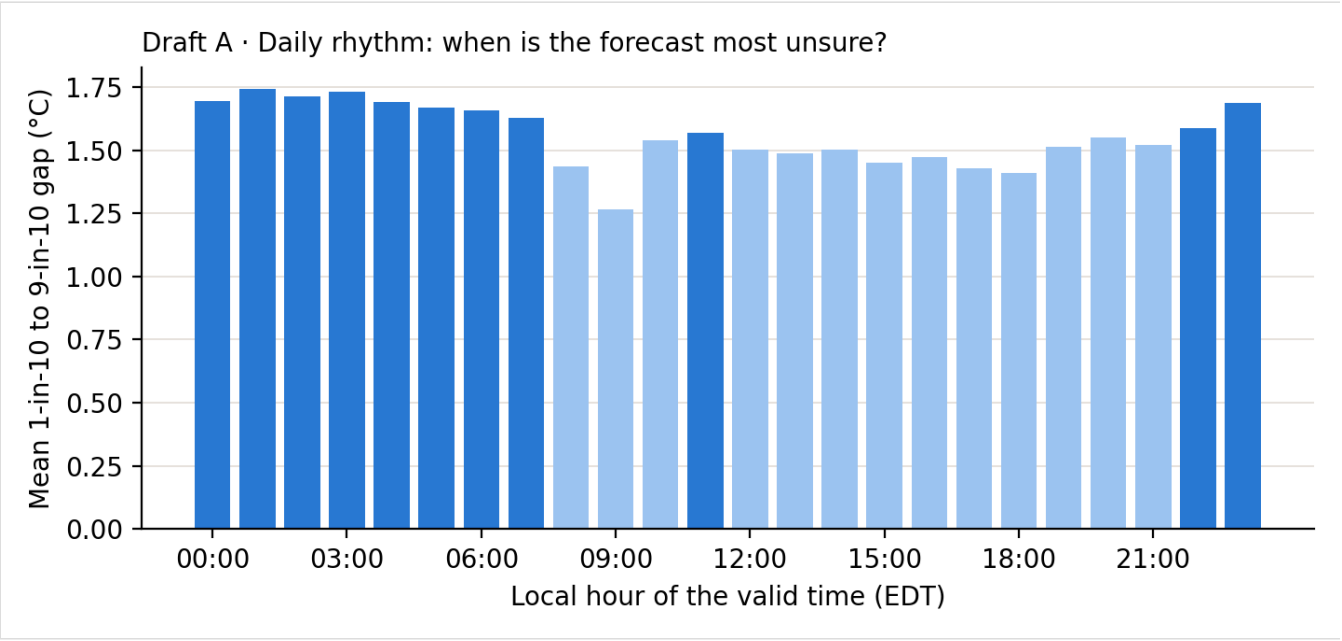


Figure 2. Mean gap by local hour, first 48 h, 12 towns. Bars at or above the average are darker.

Question: at what time of day is the forecast most or least uncertain? **Design:** a 24-hour dial (or bar strip) that sits next to the rising-insights chart. Hovering an hour shows the gap at that hour for each day. **Why it matters:** it answers the storyboard's "daily pattern" claim directly and tells a user when to check a forecast. **Open question:** the swing is only about half a degree, so the dial might need to stay subtle to avoid overstating it.

Draft B · Forecast jumps

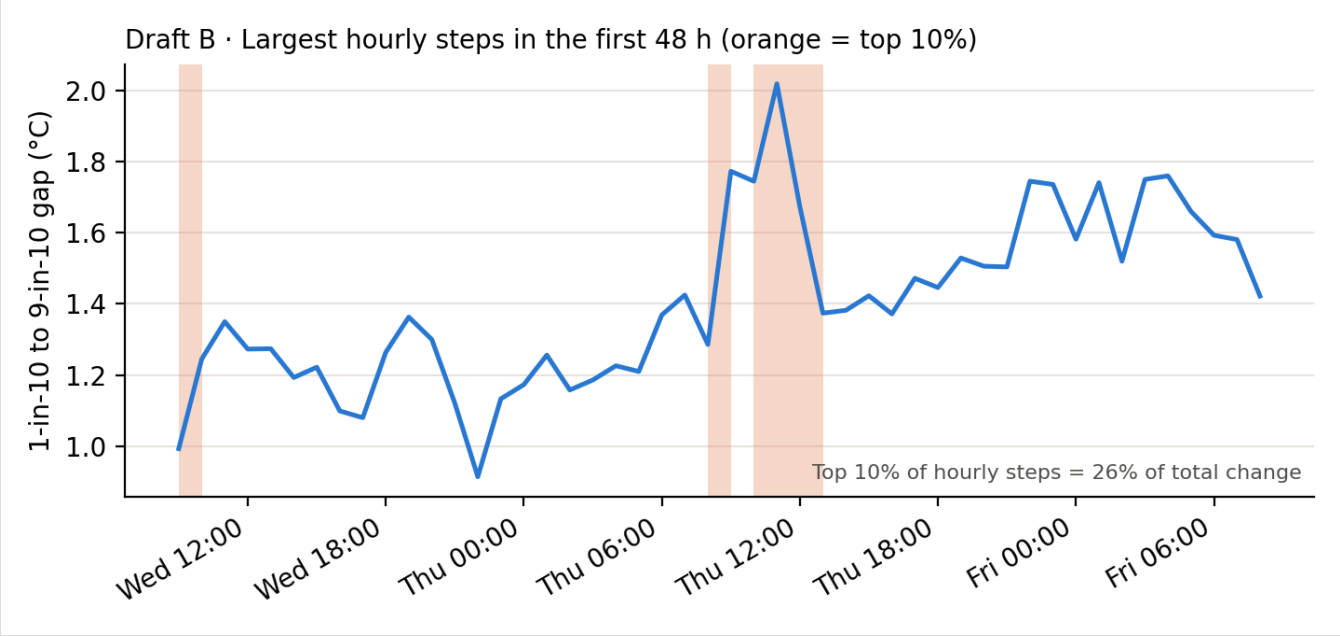


Figure 3. Hourly gap for Boston over the first 48 h. Orange spans are the 10% largest hourly steps.

Question: is uncertainty changing smoothly or in jumps? **Design:** the growth line, with its largest steps shaded and the share of change they account for in the corner. A timeline of run-to-run changes from the time machine would sit below it. **Finding that shapes this design:** in the early forecast the jumps are small, so the late-week jump (Section 6) is what the design has to show.

Draft C · Caveat strip for extreme days

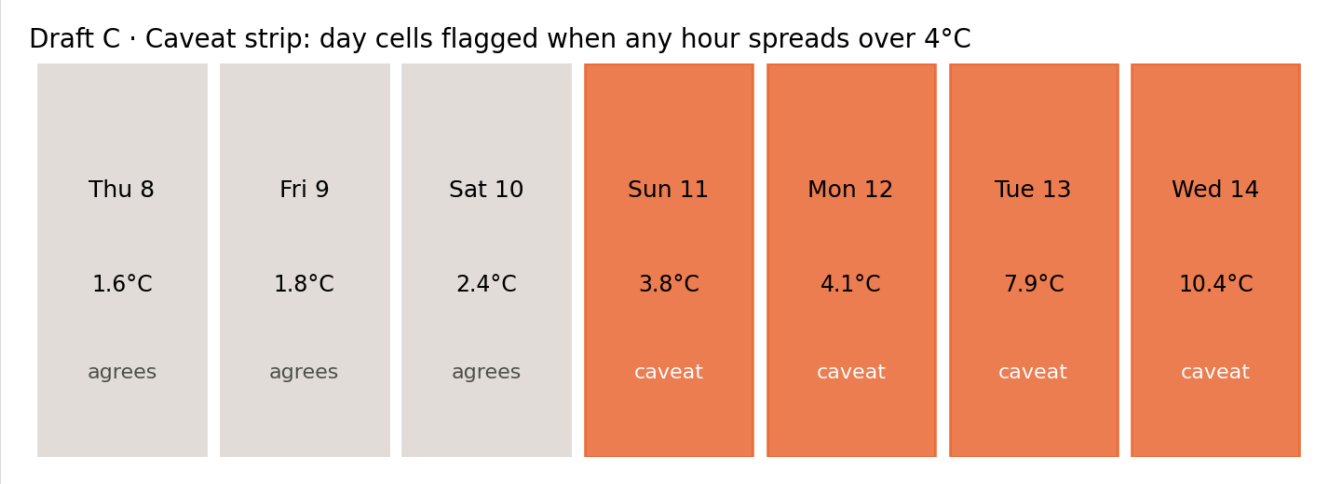


Figure 4. One cell per valid day. A cell is flagged when any hour's 10–90 gap is above 4 °C (the trust limit in the manifest).

Question: which days need a caveat? **Design:** a seven-cell strip above the planner. Flagged days show a "caveat" label and the forecast gap. The storyboard's "keyword usually, show caveats on

extreme days" becomes a visible rule instead of a footnote. **Finding:** in this run, every day from Sun 11 on has at least one hour above 4 °C, and Wed 14 is entirely above it.

8. Interaction design notes

- **Hook view:** pointer scrub for time, tap or Enter to cycle answers. Keyboard users can step through the answers; an arrow-key time scrub is planned for V2.
- **Narrative cues:** each act's heading states a claim, and each view's title is a question. Act 4 gives a rule of thumb by lead time.
- **Shared town selector:** the hook and the Act 2 views use the same place names, so a reader's town carries through the story.

9. Limitations and next steps (V2)

- One forecast run only (2026-10-07 12Z). The jumps and daily pattern need several runs before they are claims rather than observations. The time machine view has the other runs.
- Percentiles, not members. The fan and the caveat strip are built on percentiles, which is correct for the statistics we have, but the storyboard's "two far-end members" would need the member-level data.
- Confirm the coast-vs-inland result with more towns and runs before it appears as a claim.
- Add the arrow-key time scrub to the hook, and check the phone layout at a true 390 px width.
- Build Draft C into the planner page first; it is the smallest change with the clearest story payoff.